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Supplementation of the probiotic *Lactobacillus rhamnosus* GG to pre-weaning calves decreases diarrhea incidence by modulating gut bacteria and associated metabolites



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ABSTRACT

The aim of the present study was to investigate whether *Lactobacillus rhamnosus* GG (LGG) could be utilized as an effective strategy to promote growth performance of calves and to further study its impact on decreasing diarrhea. A total of 18 healthy Holstein calves (age 8 ± 3 days, weight 42.48 ± 1.04 kg) were raised in a centralized manner to minimize differences and then divided into two groups: one group received LGG supplementation (1×10^{10} CFU per calf per day), while the other served as the control (CON). Each group consisted of 9 replicates, with 1 calf per replicate. The experimental period spanned 6 weeks. The experimental results demonstrated an increase of the average daily growth ($P = 0.094$) and a highly significant reduction in diarrhea ($P = 0.001$) for the LGG group compared with the CON group. The supplementation of LGG resulted in an increased abundance of endogenous beneficial bacteria in the gut including *Lysinibacillus* ($P = 0.012$) and *Rikenellaceae_RC9_gut_group* ($P = 0.041$) while reducing levels of pathogenic bacteria in the *Escherichia-Shigella* ($P = 0.066$). These particular groups could serve as biomarker bacteria for calf diarrhea and gut health. The alteration of metabolite production and metabolic pathways were also closely associated with the variation of bacterial composition in the gut microbiome of calves, of which included enrichment of biosynthesis of unsaturated fatty acids, α -linolenic acid and steroid biosynthesis. Overall, LGG administration had a positive impact on improving the intestinal homeostasis, alleviating diarrhea, and enhancing growth performance by modulating gut bacteria and metabolites in pre-weaning calves. Thus, LGG could be a potential alternative to prophylactic antibiotic treatment that would lower associated costs and provide new insights into preventing and treating calf diarrhea.

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1. Introduction

The digestive and immune systems of calves are not fully mature, and commercial rearing and weaning often lead to diarrhea, resulting in dehydration and intestinal inflammation (He et al., 2022; Hulbert and Moisa, 2016; Klein-Jöbstl et al., 2014). Importantly, the mortality from diarrhea of calves is high, reaching 17% in Germany and 5% in the United States (Meganck et al., 2015; Renaud et al., 2019; Urie et al., 2018; Tautenhahn et al., 2020). Antibiotics have been used widely for the prevention and treatment of calf diarrhea, although long-term use can have negative effects, including the development of drug-resistant bacterial

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pathogens (Mathew et al., 2007; Bi et al., 2017). In addition, intestinal colonization by pathogenic bacteria can cause dysbiosis and weakened immune function (Ubeda and Pamer, 2012; Buffie and Pamer, 2013; Sullivan et al., 2001). Therefore, the use of alternatives to prophylactic antibiotics in farm setting, while improving gut health and preventing diarrhea, is necessary for the sustainable and healthy development of the dairy industry.

Probiotics are live bacterial inoculate that can establish in the intestinal tract and are beneficial for gut health when consumed in sufficient quantities (Zareie et al., 2006). Such as lactic acid bacteria can alleviate symptoms of gut disease via regulation of the gut microbiome and associated microbial metabolites, ultimately stabilizing the gut mucosal barrier and modulating gut mucosal immunity (Dodd et al., 2017; Liu et al., 2020; Wang et al., 2015; Rao and Samak, 2013). Probiotics have been used successfully in treating diarrheal disease and could effectively prevent diarrhea in calves when added to their diet (Cangiano et al., 2020; Renaud et al., 2019). However, one obstacle to this treatment is the action of bile and stomach acid, which can directly kill the administered bacteria. The bacteria that can survive under these conditions have been discovered, including *Lactobacillus rhamnosus* GG (LGG) with strong adhesion and resistance properties to gastric and bile acids. LGG has been tested in the human intestine and its survival period exceeds two weeks, which is sufficient to decrease the level of gut pathogens (Amdekar and Singh, 2016; Riehl et al., 2019; Zhang et al., 2019b). LGG has also been demonstrated to augment intestinal host defenses in the developing intestine by stimulating anti-apoptotic and cytoprotective responses in mice (Lin et al., 2008). In a nude mouse model, LGG was able to protect mice against the intestinal destructive effects of deoxynivalenol via promoting butyrate production from the endogenous gut microbiome and reversed gut barrier damage that resulted in increased body weights in these animals (Lin et al., 2020). LGG has been widely used in the treatment of human infantile diarrhea although the mechanism for this effect is unclear (Nixon et al., 2012).

In a previous study by Zhang et al. (2019b), feeding LGG to pre-weaning calves had a positive affection on feed intake, growth performance, rumen fermentation and decreased the incidence of calf diarrhea. Therefore, dietary supplementation with LGG may represent an effective strategy for treating diarrhea and improving survival rates in calves. In the current study, healthy calves were selected as experimental animals to further evaluate the potential role of LGG in resisting calf diarrhea and modulating gut bacteria and metabolites.

2. Materials and methods

2.1. Animal ethics statement

The animal experiments were approved by the Animal Ethics Committee of Henan Agricultural University (Permit Number: HNND2021031018) and performed in the Experiment Center of Henan Agricultural University (Zhengzhou, China). This experiment followed the Regulations of the People's Republic of China on the Administration of Laboratory Animals (2017).

2.2. Bacterial strains and cultures

Lactobacillus rhamnosus GG (ATCC 53013) used in this experiment was purchased from the China Center of Industrial Culture Collection (Beijing, China) and was originally obtained from the American Type Culture Collection (Manassas, VA, USA). The lyophilized bacterial powder was cultured in MRS broth (20220105, Sinopharm Chemical Reagent Co., Ltd., Shanghai,

China) at 37 °C with shaking (SPH-103B, Shanghai Shipping Test Equipment Co., Ltd., Shanghai, China) at 180 rpm for 24 h to achieve more than 10^{10} cells/mL. The bacteria liquid was centrifuged at $3200 \times g$ at 4 °C for 10 min (Heraeus Fresco17, Thermo Fisher Scientific, Waltham, MA, USA) and the resulting bacterial pellet was washed twice with PBS (PB180327, Procell Life Science Technology Co., Ltd., Wuhan, China). Bacterial concentration was estimated by optical density measurements at 600 nm (Lahtinen et al., 2004). The LGG cultures were prepared daily and colony-forming units (CFU) were daily measured to ensure bacterial health correlated with the OD_{600 nm} readings.

2.3. Animals and experimental design

Eighteen healthy Holstein bull calves, aged 8 ± 3 days with an average body weight of 42.48 ± 1.04 kg, were used in this study. They were centrally raised until reaching 21 days of age to minimize differences, at which point their average weight was 48.61 ± 1.03 kg. The 18 calves were randomly divided into two groups under a random block experiment design with 9 calves in the control group (CON) and 9 calves in the treatment group (LGG) that were given daily aliquots of 1×10^{10} CFU/day. All calves were raised alone on calf island for six weeks and had free access to starter feed and clean water. They were fed with a milk bottle three times a day, with each 2 L mixed with milk replacer to water ratio of 1:8, where the milk replacer was mixed thoroughly with warm water at around 50 °C, and when the temperature of the liquid milk reached about 40 °C, LGG was added and mixed well before being portioned into milk containers for feeding the calves. The composition and nutrient levels of the starter feed and milk replacer are detailed in Table 1. During the experimental period, the calves' feces were scored daily. Every two weeks, the calves were weighed before the morning feeding, and samples of feces were collected from the rectum. These samples were then stored in liquid nitrogen for determination of volatile fatty acids (VFA) and bacterial taxonomic identification.

Table 1
Composition and nutrient level of starters and milk replacer (dry matter, %).

Item	Milk replacer	Starters
Ingredients		
Corn		49.00
Soybean meal		20.00
Wheat middling		8.00
Bran		7.00
Expanded soybean		3.50
Corn gluten meal		4.00
Corn germ meal		5.00
Premix ¹		3.50
Total		100.00
Nutrition level²		
Dry matter	95.61	87.31
Organic matter	94.49	93.87
Crude protein	25.21	23.82
Ether extract	16.66	3.43
Neutral detergent fiber	4.02	25.57
Acid detergent fiber	2.11	11.74
Ca	0.47	0.91
P	0.27	0.51
Net energy ³ , MJ/kg	8.08	1.88

¹ The premix provided per kg of the starter was as follows: VA 12,000 to 14,500 IU, VD 3000 to 3500 IU, VE > 250 IU, Cu 20 to 35 mg, Fe 100 to 150 mg, Mn 40 to 60 mg, Zn 60 to 80 mg, Se > 0.7 mg, Co > 0.9 mg.

² Nutrient levels of dry matter, organic matter, crude protein; ether extract, neutral detergent fiber, acid detergent fiber, Ca and P were analyzed values.

³ The net energy value of the milk replacer was provided by the supplier, while the net energy of the starters was calculated based on the National Research Council (NRC, 2001).

2.4. Health monitoring and growth measurement

During the experimental period, the feces of all calves were scored at morning before feeding each day (0–3 points). Calves were weighed every two weeks during the experiment and the daily feed intake of each calf was recorded until the end of the experiment. The formula for calculating the diarrhea incidence of calves was as follows:

$$\text{Diarrhea incidence} = 100 \times \frac{\text{Number of calves with diarrhea}}{\text{Number of experiment calves}}.$$

2.5. Determination of feed nutrients

The nutritional levels of milk replacer and starter diets were analyzed according to the standard methods of Association of Official Analytical Chemicals (AOAC, 2012). The dry matter (DM) content was determined by drying triplicate samples at 105 °C to a constant weight according to the method 930.15. The crude protein (CP) content ($N \times 6.25$) was analyzed using the Kjeldahl apparatus (Kjeltec 2300, FOSS Analytical AB, Hoganas, Sweden) in accordance with the method 948.13. The ether extract (EE) content was quantified by a Soxhlet extractor using the method 920.39 (FOSS Soxtec8000, FOSS Analytical A/S, Hillerød, Denmark). The neutral detergent fiber (NDF) and acid detergent fiber (ADF) contents were determined via filter bag technology according to Van Soest et al. (1991) (ANKOM 2000; Ankom Technology Corp., Fairport, NY, USA). The calcium (Ca) and phosphorus (P) contents were determined via the disodium ethylenediaminetetraacetate complexometric titration method and ammonium vanadate molybdate colorimetric method by the method 968.31 and 965.17, respectively.

2.6. Determination of fecal VFA levels

The concentrations of VFA in feces were analyzed using a Sykam S-150 ion chromatograph (S-150, Sykam GmbH, Fürth, Bavaria, Germany). In brief, 1 g feces were placed in a 10-mL centrifuge tube and mixed with 4 mL ultrapure water by vortexing followed by centrifugation for 10 min at 4000 × g (Heraeus Fresco17, Thermo Fisher Scientific, Waltham, MA, USA). A 1 mL sample of the supernatant was combined with 0.2 mL of 25% metaphosphoric acid (20220125, Sinopharm Chemical Reagent Co., Ltd., Shanghai, China) in a 2-mL tube that was shaken and allowed to stand 30 min at 4 °C and centrifuged 10 min at 10,000 × g. Supernatant samples (10 µL) were combined with 990 µL ultrapure water and mixed and filtered through a 0.22-µm sterile membrane and this was used as the material for analysis. The working conditions of the ion chromatograph were as follows: column temperature of 30 °C, flow rate 1 mL/min and injection volume 10 µL.

2.7. Gut bacterial community analysis

Total bacteria DNA extraction was performed using the HiPure Soil DNA kit (D3144, Guangzhou Magen Biotechnology, Guangzhou, China) according to the manufacturers protocol. The V₃-V₄ region of 16S rDNA gene was amplified using primers 341F (5'-ACTCC-TACGGGAGGCAGCA-3') and 806R (5'-GGACTACHVGGGTWCTAAT-3') (Fei et al., 2017; Jiang et al., 2017). Amplicons were collected from 2% agarose gels and purified using the AxyPrep DNA Gel Extraction Kit (AP-GX-250, Axygen Biosciences, Union City, CA, USA) in accordance with the manufacturer's instructions and

further purified using the PCR (ABI StepOnePlus Real-Time PCR System, Thermo Fisher Scientific, Waltham, MA, USA) for quantification. The purified amplicons were paired-end sequenced on the Illumina platform (NovaSeq X, San Diego, CA, USA) according to standard procedures.

With a minimal overlap of 10-bp and 2% mismatch error rates, raw data was filtered using fast and accurate splicing tool for poly (FASTP, version 0.18.0) and paired-end clean reads were combined as raw tags using Macromedia Flash (version 1.2.11). Low-quality tags were filtered (Bokulich et al., 2013) and high-quality sequences were clustered into operational taxonomic units (OTU) with > 97% similarity. Representative sequences were classified into organisms with a naive Bayesian model using the ribosomal database project classifier (RDP, version 2.2) based on the SILVA ribosomal RNA gene database (version 132) using a confidence threshold of 0.8 to 1. The raw reads were deposited into the national center for biotechnology information (NCBI) sequence read archive database (number: PRJNA1005776).

2.8. Gut metabolite profiling analysis

Ultra-performance liquid chromatography coupled to triple quadrupole time-of-flight mass spectrometry (UPLC-Q-TOF-MS/MS, TSQ Quantum Ultra, Thermo Fisher Scientific, Waltham, MA, USA) was used to analyze gut metabolites. In brief, 50 mg feces samples were mixed with 1000 µL methanol (2021102, Tianjin Fuyu Fine Chemical Co., Ltd., Tianjin, China) and centrifuged at 12,000 × g for 10 min at 4 °C (Heraeus Fresco17, Thermo Fisher Scientific, Waltham, MA, USA). The sample supernatant was concentrated in a vacuum until dry and the residue was dissolved in 150 µL of 2-chlorobenzalanine (20220201, Sinopharm Chemical Reagent Co., Ltd., Shanghai, China) and the supernatant filtered through a 0.22-µm membrane to prepare samples for LC-MS. A 20-µL sample from each sample was used for the quality control (QC) samples and the remaining samples were used for LC-MS detection. Chromatographic separation was carried out in an Ultimate 3000 system (Ultimate 3000, Thermo Fisher Scientific, Waltham, MA, USA) using an Acquity UPLC equipped with an HSS T3 column (150 mm × 2.1 mm × 1.8 µm, Waters, Burlington, MA, USA) maintained at 40 °C with a flow rate of 0.25 mL/min.

The metabolomic data were subjected to partial least squares discriminant analysis (PLS-DA) and differential metabolite screening using GeneDenovo (<http://www.genemed.tech/gene4denovo>). The false discovery rate (FDR) < 0.05 and variable importance for the projection (VIP) > 1 was used as screening criteria. The Kyoto Encyclopedia of Genes and Genomes (KEGG) database was applied to functionally annotate differential metabolites and Spearman correlation analysis was used to determine correlations between different feces bacteria and metabolites in feces, respectively.

2.9. Statistical analysis

Calf body weights, feed intake, feces score and diarrheal incidence were preprocessed using Microsoft Excel 2019. SPSS 26.0 (IBM, Chicago, IL, USA) software to assess variance using Levene's test and independent sample *t*-tests were subsequently applied to evaluate statistical significance. The formula for the *t*-test analysis model is as follows:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}},$$

$$S_p = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}$$

where S_p is the pooled standard deviation, $\bar{x}_1$ and $\bar{x}_2$ are the means of the two sample groups, respectively; S_1^2 and S_2^2 are the variances of the two sample groups, respectively; and n_1 , n_2 are the sample sizes of the two groups, respectively. The variation of indexes was expressed as the standard error of means (SEM). $P < 0.05$ was defined as statistical significance and $0.05 < P < 0.10$ was defined as a trend of difference. Graphs were drawn using GraphPad Prism 8.0 (San Diego, CA, USA) software.

3. Results

3.1. Animal growth measurements

All calves received an equal amount of milk replacer throughout the trial and the starter was offered ad libitum. Feed intake and calf growth measurements indicated an increasing trend of average daily growth (ADG) of calf supplementation with LGG at 0 to 6 weeks ($P = 0.094$). The dry matter intake of calves exhibited no significant difference between the CON and LGG group calves (Table 2). In addition, there was no significant difference in feces scores between LGG group and CON group throughout the experiment ($P = 0.734$). However, the incidence of diarrhea in the LGG group calves was lower than CON calves for the whole experiment period ($P = 0.001$) (Table 3).

3.2. Volatile fatty acids measurements

The concentration of acetic acid in calf feces gradually increased with the increase of feeding time, and the acetic acid concentration in the LGG group was substantially higher than that of CON calves ($P = 0.005$). The concentration of butanoic acid in feces of LGG group increased gradually with time but the same increased trend was not found in the CON group. The levels of propionic acid ($P = 0.001$), butanoic acid ($P = 0.002$) and total VFA ($P = 0.002$) in the LGG group were significantly higher than that of CON group by 6 weeks. No significant differences were found in isobutanoic acid, valeric acid or isovaleric acid concentrations between the two groups during the six-week experimental period (Table 4).

3.3. Effect of LGG on the calf gut bacteria composition

The VFA alterations between two groups of calves indicated that LGG had influenced the gut microbiome of these animals.

Table 2
Effects of feeding LGG on calf feed intake and growth performance.

Item	Times	Treatment		SEM	P-value
		CON	LGG		
Body weight, kg	0 week	48.6	48.6	0.98	0.999
	2 week	56.8	56.3	1.04	0.789
	4 week	67.5	67.9	1.34	0.877
	6 week	79.2	82.8	1.75	0.321
Average daily gain, g/d	0–2 week	579.8	584.8	25.72	0.927
	3–4 week	761.9	872.5	48.33	0.271
	5–6 week	837.3	1122.5	75.02	0.056
	0–6 week	728.8	854.0	26.62	0.094
Dry matter intake, g/d	0–2 week	915.2	883.0	15.66	0.327
	3–4 week	1130.1	1148.5	35.09	0.802
	5–6 week	1690.7	1729.0	66.63	0.783
	0–6 week	1245.4	1253.7	37.68	0.916

CON = control group ($n = 9$); LGG = fed the *Lactobacillus rhamnosus* GG group ($n = 9$); SEM = standard error of the mean.

Table 3
Effects of feeding LGG on calf feces scores and diarrhea incidence.

Item	Times	Treatment		SEM	P-value
		CON	LGG		
Feces score (0–3)	0–2 week	1.3	1.2	0.06	0.538
	3–4 week	1.1	1.0	0.04	0.524
	5–6 week	1.2	1.1	0.03	0.125
	0–6 week	1.1	1.1	0.03	0.734
Diarrhea incidence, %	0–2 week	16.7 ^a	6.4 ^b	2.59	0.042
	3–4 week	7.9 ^a	2.4 ^b	1.27	0.023
	5–6 week	7.9	0.8	2.01	0.075
	0–6 week	10.8 ^a	3.2 ^b	1.30	0.001

CON = control group ($n = 9$); LGG = fed the *Lactobacillus rhamnosus* GG group ($n = 9$); SEM = standard error of the mean.
In the same row, values with different small letter superscripts designate significant differences ($P < 0.05$).

Table 4
Effects of feeding LGG on the content of volatile fatty acids in calf feces (μmol/g).

Item	Times	Treatment		SEM	P-value
		CON	LGG		
Acetic acid	2 week	24.71	28.16	1.447	0.250
	4 week	29.02	30.41	0.928	0.480
	6 week	29.44 ^b	40.08 ^a	2.213	0.005
Propionic acid	2 week	6.07 ^b	8.53 ^a	0.486	0.004
	4 week	7.63	7.43	0.228	0.674
	6 week	5.46 ^b	9.21 ^a	0.670	0.001
Butanoic acid	2 week	3.14	3.54	0.137	0.145
	4 week	3.62 ^b	4.06 ^a	0.095	0.012
	6 week	3.36 ^b	4.25 ^a	0.168	0.002
Isobutanoic acid	2 week	1.48	1.46	0.149	0.963
	4 week	0.93	0.98	0.081	0.787
	6 week	0.64	0.55	0.071	0.401
Valeric acid	2 week	1.54	1.43	0.177	0.765
	4 week	1.15	1.33	0.207	0.479
	6 week	1.37	1.15	0.158	0.513
Isovaleric acid	2 week	1.37	1.21	0.183	0.683
	4 week	0.79	0.87	0.192	0.705
	6 week	0.53	0.60	0.064	0.617
Total VFA	2 week	38.01	44.62	1.752	0.054
	4 week	43.45	44.77	1.073	0.565
	6 week	40.56 ^b	56.48 ^a	2.991	0.002

CON = control group ($n = 6$); LGG = fed the *Lactobacillus rhamnosus* GG group ($n = 6$); SEM = standard error of mean; VFA = volatile fatty acid.
In the same row, values with different small letter superscripts designate significant differences ($P < 0.05$).

Consequently, the 16S rRNA gene sequencing was employed to investigate the gut bacterial composition. The results revealed that with the extension of feeding time, the abundance and diversity of bacteria increased. But no significant difference was observed between the two groups (Fig. 1A and B). However, principal co-ordinates analysis (PCoA) plots of OTU indicated an obvious spatial separation between the CON and LGG groups and these differences were most obvious by week 6 (Fig. 1C and D). The bacterial phyla that predominated were Firmicutes, Proteobacteria, Bacteroidetes and Actinobacteria that accounted for 99% of total bacterial phyla (Fig. 2A). Proteobacteria abundance in the CON group gradually increased but decreased in the LGG group. Bacteroidetes levels for the LGG group rose gradually with time reaching 4.94%, 7.84% and 21.35% at 2, 4 and 6 weeks, respectively. The genera *Solibacillus*, *Escherichia-Shigella*, *Bacteroides*, *Rikenellaceae_RC9_gut_group*, *Psychrobacillus* and *Ruminococcaceae_UCG_005* predominated and accounted for 70% of the total (Fig. 2B). By week 2, the levels of *Paenibacillus* ($P = 0.089$), *Bacillus* ($P = 0.093$) and *Ruminococcus_gaureauii_group* ($P = 0.086$) for the LGG group were lower than that of the CON group. At week 4, *Parabacteroides* ($P = 0.017$),

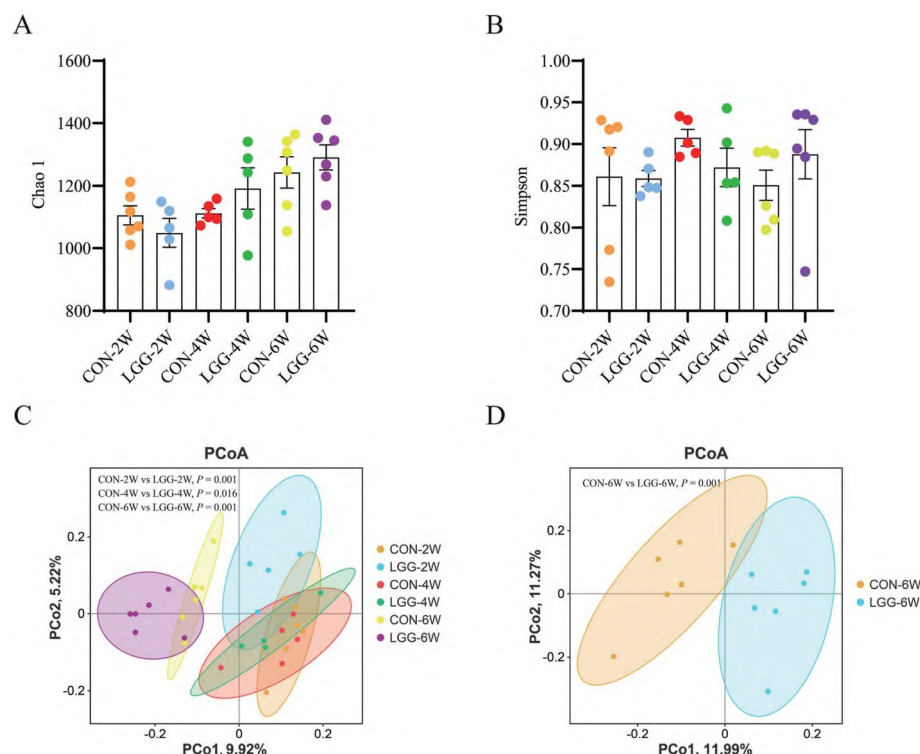


Fig. 1. Community abundance and diversity of feces sample from calves. Alpha diversity was evaluated based on the (A) Chao 1 and (B) Simpson indices using OTU abundance. (C, D) Bacterial similarity between CON and LGG groups using PCoA scatter plots as indicated. CON = control group ($n = 6$); LGG = fed the *Lactobacillus rhamnosus* GG group ($n = 6$); W = week; OTU = operational taxonomic unit; PCoA = principal component analysis.

Christensenellaceae_R_7_group ($P = 0.096$) and *Olsenella* ($P = 0.085$) were significantly higher in the LGG group compared with CON. At week 6, *Escherichia-Shigella* ($P = 0.066$) was significantly higher in the CON group while *Rikenellaceae_RC9_gut_group* ($P = 0.042$) and *Lysinibacillus* ($P = 0.012$) were significantly higher in the LGG group (Fig. 2C–E).

Linear discriminant analysis effect size (LefSe) was used to further determine whether there was a differential enrichment of specific bacterial communities of calves in the sixth week. With a logarithmic linear discriminant analysis (LDA) score cutoff set at 2.0, the experiment identified 47 discriminatory genera as key discriminants (Fig. 2F). The *Rikenellaceae_RC9_gut_group* and *Lysinibacillus* were remarkably enriched in the feces of LGG group calves whereas *Actinobacteria*, *Xanthomonadles*, *Micrococcales* and *Micrococcaceae* were dominant bacteria in CON group calves. A cladogram depicting the taxonomic hierarchy of the fecal bacteria from the phylum to genus levels revealed a notable difference in the phylogenetic distribution of the bacteria between the calves in LGG and CON groups by week 6 (Fig. 2G).

To identify the potential interactions and niche-sharing among bacteria, the experiment constructed gut bacteria co-occurrence and mutual exclusion networks related to the CON and LGG groups calves at the sixth week (Fig. 3A). A spearman's correlation between two bacteria with $|R| > 0.5$, $P < 0.05$ was indicative of a positive linkage. The experiment found that the genera *Ruminococcaceae_UCG_005*, *Phascolarctobacterium*, *Cellulosimicrobium*, *Christensenellaceae_R_7_group* and *Escherichia-Shigella* were the primary members of the network diagram. The experiment further found that *Solibacillus* ($R = -0.557$, $P = 0.001$) and *Lysinibacillus* ($R = -0.727$, $P = 0.010$) were negatively correlated with *Escherichia-Shigella* (Fig. 3B–E).

3.4. Effects of LGG on the calf gut metabolic profiling

In the current study, feces metabolites were detected using untargeted metabolomics to systematically map interactions between the gut bacteria and their associated metabolites. PCoA demonstrated that the QC samples were closely clustered indicating a stable experimental system (Fig. S1A). Metabolite identification revealed 6689 and 5443 peaks in positive and negative ion modes, respectively. A PLS-DA plot also displayed significant separations of gut metabolites between the LGG and CON groups across the whole experiment (Fig. 4A, Fig. S1B and S2B). Subsequently, univariate statistical analysis of the t -test and VIP values of PLS-DA was used to screen differential metabolites between LGG and CON groups ($VIP > 1$, $P < 0.05$). A total of 335 differential metabolites were screened between the CON and LGG groups that included 101 metabolites in week 2 (Fig. S1C and D) and 55 in week 4 (Fig. S2C and D). In the sixth week, the experiment pinpointed 80 metabolites that were up-regulated, with zerumbone and terephthalic acid among them. Additionally, 79 metabolites were identified as down-regulated, such as zearalenone and cycloheximide (Fig. 4B and C).

All differential metabolites were functionally annotated using KEGG and 20 metabolic pathways were significantly altered after feeding LGG. These included lipid metabolism (α -linolenic acid metabolism, steroid and unsaturated fatty acid biosynthesis and glycerolipid metabolism), amino acid metabolism (glycine, serine and threonine metabolism and cysteine and methionine metabolism), metabolism of other amino acids (selenocompound metabolism, phosphonate and phosphinate metabolism), xenobiotic biodegradation and metabolism (steroid degradation, aminobenzoate degradation, dioxin degradation), carbohydrate

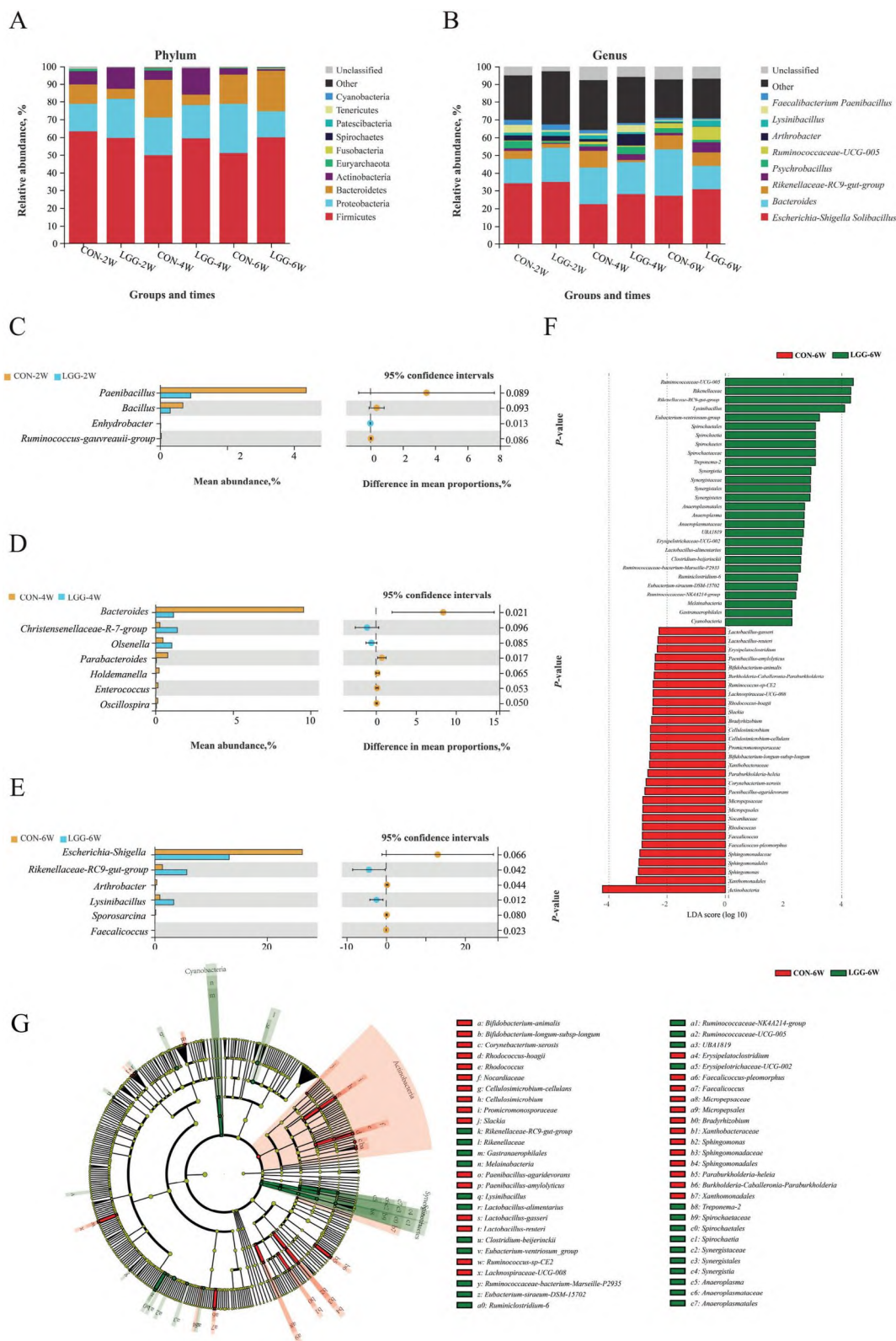


Fig. 2. Gut bacteria composition spectrums in calves. Taxonomic distributions of calf fecal bacteria at the (A) phylum and (B) genus levels (top 10). (C, D and E) Statistical analysis using the Wilcoxon rank-sum test of genus-level bacteria that significantly differed between CON and LGG group calves. (F) LDA score histograms were used to identify bacterial genera that differed significantly between LGG and CON groups calves in the sixth week (LDA score >2). (G) Microflora distribution of feces bacteria from phylum to genus levels in sixth week in LGG and CON groups calves as indicated using LefSe. CON = control group (n = 6); LGG = fed the *Lactobacillus rhamnosus* GG group (n = 6); W = week; LDA = linear discriminant analysis; LefSe = linear discriminant analysis effect size.

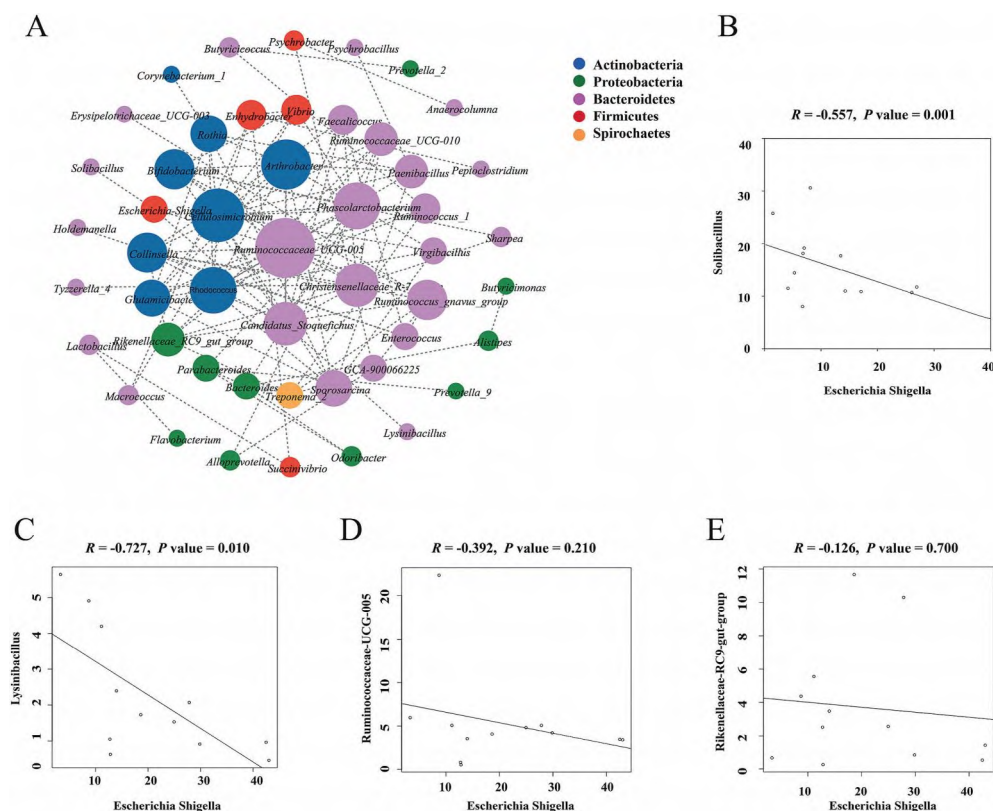


Fig. 3. Gut bacteria interaction networks. (A) CON and LGG groups calves gut bacteria co-occurrence network at week 6. Relationships between the orders (B) *Solibacillus*, (C) *Lysinibacillus*, (D) *Rikenellaceae_RC9_gut_group*, and (E) *Ruminococcaceae_UCG_005* with *Escherichia-Shigella* in the gut. CON = control group ($n = 6$); LGG = fed the *Lactobacillus rhamnosus* GG group ($n = 6$); W = week.

metabolism (starch and sucrose metabolism), energy metabolism (carbon fixation in photosynthetic organisms) and drug resistance (vancomycin resistance) (Fig. 4D).

3.5. Correlation between differential gut bacteria and metabolites

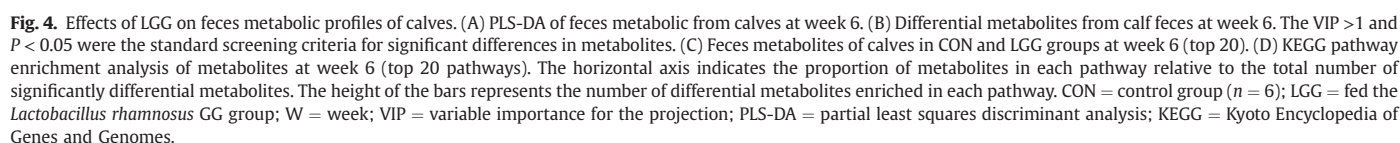
The results of bacteria and metabolites indicated significant differences in the gut microbiomes of calves between the CON and LGG groups. To further clarify the association between gut bacteria and metabolites, Spearman's correlation analysis was performed and indicated an association of gut bacteria with VFA and incidence of diarrhea. Acetic acid was positively correlated with the relative abundance of *Rikenellaceae_RC9_gut_group* ($P = 0.009$), *Christensenellaceae_R_7_group* ($P = 0.010$), *Ruminococcaceae_UCG_005* ($P = 0.002$), *Ruminococcaceae_UCG_013* ($P = 0.002$), *Alloprevotella* ($P = 0.025$) and *Phascolarctobacterium* ($P = 0.003$). Negative correlations were found with the relative abundance of *Arthrobacter* ($P = 0.049$), *Faecalibacterium* ($P = 0.010$), *Bifidobacterium* ($P = 0.005$) and *Collinsella* ($P = 0.021$). The concentration of propionic acid was positively correlated with that of *Lysinibacillus* ($P = 0.010$) and negatively correlated with *Psychrobacillus* ($P = 0.003$). Isobutanoic acid was positively correlated with *Arthrobacter* ($P = 0.004$), *Bifidobacterium* ($P = 0.001$) and negatively correlated with *Ruminococcaceae_UCG_005* ($P = 0.002$), *Phascolarctobacterium* ($P = 0.001$) and others. Additionally, the relative abundance of *Ruminococcaceae_UCG_013* was negatively correlated with the incidence of diarrhea ($P = 0.024$) (Fig. 5A).

The experiment further investigated associations between gut differential metabolites and bacteria. The relative abundance of *Phascolarctobacterium*, *Ruminococcaceae_UCG_005*, *Christensenellaceae_R_7_group*, *Rikenellaceae_RC9_gut_group*,

Candidatus_Stoquefichus, *Ruminococcaceae_UCG_010* and *Ruminococcaceae_UCG_013* were positively correlated with citalopram, isoliquiritigenin, phenobarbital, difenacoum, chlorogenic acid, linolenic acid and monocrotaline ($P < 0.05$) but negatively correlated with biphenylindanone A, altanserine and arbutin ($P < 0.05$). The relative abundance of *Arthrobacter*, *Collinsella*, *Glutamicibacter* and *Ruminococcus_gnavus_group* were negatively correlated with phenobarbital, difenacoum, chlorogenic acid, linolenic acid and monocrotaline ($P < 0.05$) and positively correlated with biphenylindanone A and arbutin ($P < 0.05$). Moreover, ursodeoxycholic acid was negatively correlated with *Escherichia-Shigella* ($P = 0.006$) (Fig. 5B).

4. Discussion

The cattle industry has sustained serious economic loss as result of calf diarrhea and the use of prophylactic antibiotics to treat this problem has become unsustainable. Drug resistant pathogens also pose a threat to human health even though antibiotics had previously been effective in preventing diarrhea in calves. In contrast, there is a body of evidence for a beneficial role of probiotics in modulating gut health and improving growth performance in animals. For example, *Lactobacillus casei*, *Lactobacillus animalis* LA51 and *Propionibacterium freudenreichii* PF24 have positive effects on reducing calf diarrhea and improving growth performance (Stefańska et al., 2021). Furthermore, LGG has been utilized to prevent and treat human and animal diarrhea by regulating gut microflora and improving gut mucosal immunity (Barai et al., 2018; Li et al., 2019; Tang et al., 2019). The current study found the incidence of diarrhea was reduced and the calf ADG had the tendency



A prominent role in maintaining gut health is VFA production by gut bacteria that act to modulate gut barrier function and improve gut immunity (Gagné et al., 2013; Blaak et al., 2020). The experiment found that when LGG was given to the calves, the levels of acetic, propionic and butanoic acids in the gut increased significantly. This increase is most likely the reason for the significantly lower level of diarrhea for the LGG group. VFA production was conducive to LGG colonization and this also promoted endogenous probiotics such as the lactic acid bacteria to flourish. As an added benefit, these compounds are inhibitory to pathogenic genera such as *Salmonella* (Van Immerseel et al., 2004). In this study, the experiment found that feeding LGG to calves improved the activity and abundance of VFA-producing bacteria in the calves' gut and elevated levels of key organic acids that have a positive effect on gut health.

Diarrhea and gut inflammation are the result of interactions between the external environment, gut bacteria and host immunity (Citi, 2018; Kostic et al., 2014; Round and Mazmanian, 2009; Halfvarson et al., 2017). Bacteria gradually colonize animals due to their initial contact with food and the environment (Virgínio Júnior and Bittar, 2021; Du et al., 2023). In the current study, bacteria diversity increased over time indicating a gradual colonization in the gut. The relative abundance of Proteobacteria decreased with time for the LGG group but increased in the CON group. Proteobacteria are gram-negative bacteria and are a marker of gut microbial imbalance and intestinal inflammation (Henderson and Lam, 2001; Shin et al., 2015). Therefore, LGG has a positive impact on improving gut health most likely due to reductions in the levels of Proteobacteria. The experiment also revealed that in the LGG group calves, the abundances of the beneficial genera *Lysin-bacillus* and *Rikenellaceae_RC9_gut_group* increased, while the abundance of the pathogen *Escherichia-Shigella* decreased.

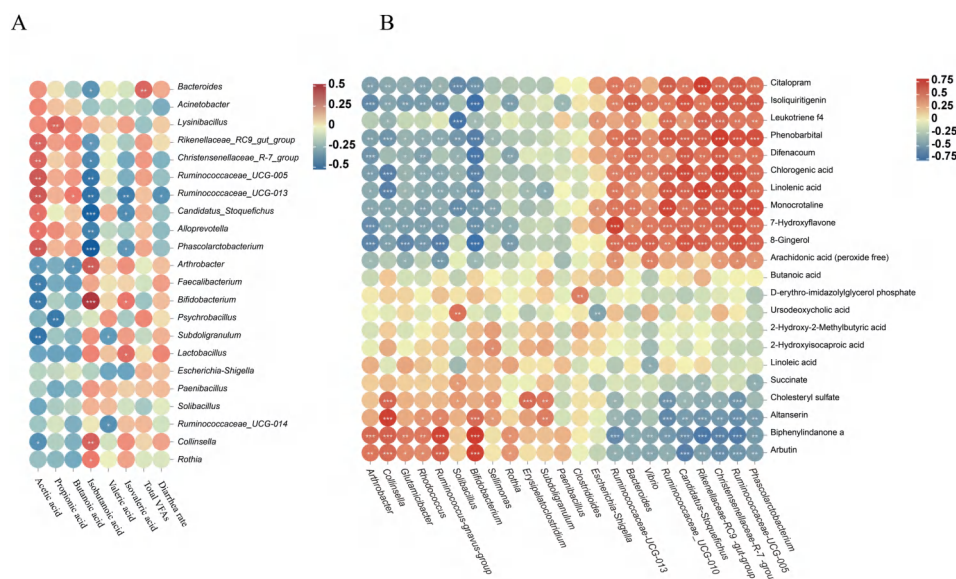


Fig. 5. Relationships between (A) VFA and bacteria abundance and (B) Gut bacteria and metabolites. R values were displayed in different colors. *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$. The legend on the right is the color interval of different R values. VFA = volatile fatty acid.

Consistent with this, our network results demonstrated a negative correlation between *Lysinibacillus* and *Escherichia-Shigella*. The latter group target and adhere to the intestinal mucosal epithelium and are recognized to cause diarrhea and dysentery in animals (Yan et al., 2022; Zhang et al., 2020). The increase in *Lysinibacillus* abundance and the decrease in *Escherichia-Shigella* abundance are most likely direct reasons for relieving calf diarrhea and improving growth performance. *Escherichia-Shigella* have been directly implicated in diarrhea in calves (Deng et al., 2022). In addition, the presence of *Lactobacillus gasseri* could reduce calf diarrhea caused by *Escherichia coli* and reduce the relative abundance of *Escherichia-Shigella* in the gut of these animals (Yue et al., 2020). These data indicated that *Escherichia-Shigella* and *Lysinibacillus* could be used as biomarker bacteria to evaluate calf diarrhea and calf gut health.

The metabolites of gut bacteria are closely related to the host physiological state that includes nutritional and health status and stress responses (Sekirov et al., 2010; Yang et al., 2022). In the current study, the experiment results indicated that some metabolites differed significantly between the LGG and CON groups. These differential metabolites were primarily enriched for α -linolenic and arachidonic acid metabolism, steroid and unsaturated fatty acid biosynthesis and steroid degradation. Arachidonic acid is a key precursor of prostaglandin synthesis can be advantageous for exerting antifungal, antiviral and immunomodulatory effects (Das, 2018; Naito et al., 2015). The essential α -linolenic acid plays an important role in regulating blood lipids, is anti-microbial, protects vision and acts to balance the gut microflora and modulate gut inflammation (Ni et al., 2021; Mizuno and Taguchi, 2020; Kaveh et al., 2019) and is inhibitory to *Escherichia-Shigella* and *Staphylococcus aureus* (Leat et al., 1986; Giamarellos-Bourboulis et al., 1995; Lee et al., 2002). Thus, α -linolenic acid enrichment in the calf gut can reduce the growth and colonization of pathogenic bacteria such as *Escherichia-Shigella* and counteract their ability to cause calf diarrhea. Other compounds may also be used as adjuvants in addition to LGG therapy. For instance, the algal and rape plant compound brassicasterol has a positive effect on immunity and cardiovascular and colon health and its content addition could improve calf immunity and anti-inflammatory ability (Hu et al., 2017; Bouic, 2001). Our data are also consistent with studies that indicated that diarrhea progression in calves in highly correlated

with the presence or absence of particular gut metabolites and directly implicates the gut microbiome in maintaining gut health (Dodd et al., 2017; Xu et al., 2022; An et al., 2021).

A correlation analysis between bacteria and metabolite variations also revealed that α -linolenic and arachidonic acid pathways were positively correlated with *Phascolarctobacterium* and *Ruminococcaceae_UCG_005*, *Ruminococcaceae_UCG_010* and *Ruminococcaceae_UCG_013*. This linked gut bacterial populations with their metabolic phenotype. In addition, an interesting finding was a significantly negative correlation between *Escherichia-Shigella* and ursodeoxycholic acid and this has been previously reported (He et al., 2022). Ursodeoxycholic acid can improve gut integrity and inhibit *Escherichia coli* (Goossens and Bailly, 2019; He et al., 2022).

5. Conclusions

The current study results indicated that supplementing with LGG reduced the incidence of diarrhea in pre-weaning calves and dramatically altered the gut bacteria composition and associated metabolic profiles. LGG increased the abundance of health-related *Lysinibacillus* and *Rikenellaceae_RC9_gut_group*, while decreased the abundance of the pathogen *Escherichia-Shigella*. Additional positive effects of LGG have also been found, including enriched biosynthesis of unsaturated fatty acids, α -linolenic acid metabolism and steroid biosynthesis. Therefore, LGG exhibited great potential in improving the gut health and alleviating diarrhea by modulating gut bacteria and metabolites in pre-weaning calves. These findings might provide new insights into the application of probiotics.

Credit Author Statement

Haoqi Han: Writing – original draft, Methodology, Investigation, Conceptualization. **Kaizhen Liu:** Writing – review & editing, Methodology. **Yao Wang:** Visualization, Validation, Data curation. **Mingming Song:** Visualization, Validation, Data curation. **Hongxia Lian:** Supervision, Methodology. **Tengyun Gao:** Supervision, Conceptualization. **Liyang Zhang:** Writing – review & editing, Resources, Methodology, Funding acquisition, Conceptualization. **Tong Fu:** Writing – review & editing, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, and there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the content of this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.aninu.2025.02.005>.

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